

SUBJECT: Sustainable Living

SUMMARY:

Our educational game on sustainable living is tailored to diverse users from 3rd-9th grade (age 8-14) and offers a tiered curriculum that both fosters foundational awareness and inspires real-world actions. The game employs three pedagogical models – constructivism, spiral curriculum, and constructionism – to achieve its educational goals. Specifically, our curriculum ensures (1) learning topics take into account limited pre-existing knowledge of the young audience, (2) later topics build on and deepen previous knowledge, and (3) there are concrete deliverables/actions the users can take as the fruit of their learning.

The game presents three distinct learning levels. The first level, foundational awareness, acquaints users with the core elements of their environment. The second level, everyday practices, provides actionable insights that can be integrated into their daily routines. Finally, the curriculum culminates in community and systemic actions, challenging learners to engage in the broader socio-political contexts of environmentalism and sustainability, empowering them to be a green citizen and an agent of change.

TARGET USER ATTRIBUTES

User 1: An urban teen with a passion for sports and entertainment video games. He occasionally reads comics, struggles with middle school math, and has limited sustainability awareness.

- Age: 13
- Hobbies: Sports, playing video games, occasionally reading comics
- Stage of life: Teen
- Personality: Curious, somewhat tech-savvy, and introverted
- Highest level of mathematical skills: Elementary; Struggles with middle school math concepts
- Highest level of reading skills: Elementary; Prefers visual content and has difficulty with dense text
- Level of technical skills: Proficient with video game consoles but not with productive digital tools
- Devices owned: Personal smartphone, gaming console, shared family computer
- Other relevant skills: Limited knowledge of concepts related to nature, eco-friendliness, or sustainability
- Generations lived in the US: Third generation
- Cultural background: African-American
- Gender: Male
- Living area: Urban

User 2: A boy from a rural farm, familiar with sustainable practices but lacks theoretical knowledge and broader awareness of environmental issues. Uses minimal technology.

- Age: 11
- Hobbies: Fishing, exploring the woods, helping out on their family's farm
- Stage of life: Child
- Personality: Outdoorsy, independent, practical

- Highest level of mathematical skills: Average for their grade
- Highest level of reading skills: Prefer hands-on activities, reads mostly when required
- Level of technical skills: Rarely uses computers, only for school assignments
- Devices owned: Basic cell phone
- Other relevant skills: Practical knowledge of farming but not the underlying science
- Generations lived in the US: First generation (immigrant family)
- Cultural background: Hispanic
- Gender: Male
- Living setting: Rural

User 3: A suburban girl interested in nature and reading. Uses her tablet mainly for videos and has color vision deficiencies (protanopia and deuteranopia.)

- Age: 9
- Hobbies: Birdwatching, collecting stickers, playing with pets
- Stage of life: Child
- Personality: Inquisitive, energetic, social
- Highest level of mathematical skills: Above average for their grade, enjoys problem solving
- Highest level of reading skills: Enjoys children's novels
- Level of technical skills: Uses tablets to watch Youtube videos, not familiar with games or educational apps
- Devices owned: Tablet with restricted access
- Other relevant skills: In-depth knowledge of local birds but not broader environmental concepts
- Generations lived in the US: Fourth generation
- Cultural background: Asian-American
- Gender: Female
- Living setting: Suburban

SKILLS

Beginner (Foundational Awareness):

- Resource scarcity
- Ecosphere
 - Biodiversity
 - Pollution
 - Animal welfare

Intermediate (Everyday Practices):

- Upcycling
- Hand-me-down
- Trash sorting
- Energy conservation
- Public transportation
- Chemical alternatives

- Climate change

Advanced (Community and Systemic Actions):

- Circular economy
- Industrial waste accountability
- Community initiative
- Political participation
- Green labels

LEARNING TRAJECTORY DIAGRAM (attached .pdf file)

Development Process: Our methods for building a learning trajectory diagram take place in four phases:

- (1) Compiling learning topics from online resources
- (2) Filtering the topics for our target audience
- (3) Clustering topics by difficulty level
- (4) Drawing connections between topics into a coherent trajectory

Theoretical Frameworks and Our Adaptation: We refer to and adapt the following educational models and theories:

- **Spiral Curriculum:** We adapted the spiral approach by assembling learning objectives in a way that new topics are directly connected to and contextualized by what students have previously learned and gradually increase in difficulty, which not only makes new knowledge easier to absorb but also reinforces and enhances their understanding of foundational concepts.
- **Constructivism:** Topics are chosen and classified for difficulty according to alignment with the young audience's pre-existing knowledge and mental models. As such, some sustainability topics that require too complex of an academic background (and lack immediate actionability for young users—also as per constructionism) have been filtered out, such as renewable energy and urban planning. We will ensure to incorporate game mechanisms that encourage active learning activities and reflection on the learning journey itself.
- **Constructionism:** We adapt the sharable artifacts central to a constructionist curriculum as tangible real-world actions in ours. Users are encouraged to own the outcomes of their learning in a public setting.

Existing Related Materials: As a practical/non-academic curriculum, the primary objective of our game is not to impart a single, overarching learning goal but rather to instill in kids a profound sense of responsibility and preparedness for engaging in a multitude of sustainable actions. Drawing on existing learning goals and ordering, our learning trajectory is structured to encompass three levels of understanding and engagement:

- This level focuses on introducing the core elements and artifacts that make up our environment, such as resource scarcity, pollution, and animal welfare. Through this, students gain awareness of the basic elements that shape our world and why they matter. The idea is to provide them with the vocabulary and initial insights needed to participate in more advanced conversations. [Source 1, 2].

- Here, the emphasis shifts to actions that individuals can seamlessly integrate into their daily lives. These are measures that most young children are partly familiar with or can easily adopt. From upcycling to energy conservation, this level delves into simple yet impactful behaviors that contribute to a more sustainable lifestyle. It's all about taking the foundational knowledge from the beginner level and applying it in real, tangible ways. [Source 3].
- This level is designed for students ready to engage with sustainability in a broader social, political, and economic context. The actions covered here may not be as immediately familiar and generally require a more concerted effort, either mentally, physically, or both. Whether it's understanding the circular economy or participating in local politics, this level empowers students to influence systemic change. It goes beyond personal responsibility to include community involvement and institutional accountability. [Source 1, 3].

Sources:

- [1] [Where Does IT Go?](#)
- [2] [The Soil Story Curriculum](#)
- [3] [Kids Against Climate Change](#)

LEARNING TRAJECTORY DIAGRAM

(see attached Learning Trajectory Diagram document)